BUILD WITH GEMMA · GDG PWANI HACKATHON · TRACK 3: SMART HEALTH

AfyaPulse

The heartbeat Kenya's health supply chain never had stock-out forecasting, LP-optimized redistribution, and a Gemma 4 copilot a District Health Officer can act on today.

github.com/Mamiluu/Hackathon · afyapulse-blond.vercel.app · afyapulse-forecasting.onrender.com

Kenya's health system isn't blind, it's fragmented. DHIS2, eCHIS, and KEMSA's LMIS don't talk to each other, so a facility can sit at zero stock on a medicine while another an hour away holds surplus of the same item. With SHA's 90-day digitize-or-de-contract deadline now live, this is a regulatory problem, not just an operational one. AfyaPulse is the reasoning layer that sits on top of those existing systems and turns scattered stock, bed, staffing, and diagnostics data into a stock-out forecast, a redistribution plan, and a copilot a District Health Officer can act on today.

ARCHITECTURE — SENSE, REASON, TRUST

01 · SENSE

Field intake

CHV/nurse captures a stock count by photo or voice, in any of 14 languages 2 hand-verified, 12 auto-translated live by Gemma 4 itself, cached as static assets. No forms, no re-typing registers.

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02 · REASON

The district brain

Holt's linear trend forecasting (not Croston's wrong tool for non-sparse demand) predicts stock-outs. SciPy linear programming solves cross-facility transfers. Gemma 4 wraps both with native function calling.

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03 · TRUST

Human in the loop

Every number traces back to source data. Every redistribution needs an explicit "Approve & Dispatch" click. Nothing moves stock autonomously.

8 FACILITIES MONITORED	**60d** SEEDED CONSUMPTION HISTORY	**4** COPILOT FUNCTION-CALL TOOLS	**0** API KEYS NEEDED TO RUN IT

ACTUALLY RUNNING, NOT SLIDEWARE

- **District Command Center** — live health score, alerts, stock/bed/staffing/diagnostics across 8 Kilifi County facilities, with a deliberate crisis scenario baked into the seed data.
- **Stock-out forecasting** — Holt's linear trend exponential smoothing on real consumption history via a FastAPI microservice.
- **Smart redistribution** — transportation-problem LP (`SciPy.optimize.Linprog`) minimizing transit distance under each facility's 10-day safety buffer.
- **Gemma 4 DHO Copilot** — agentic chat, native function calling over 4 live-data tools.
- **Gemma 4 dispatch briefs** — one click turns a transfer proposal into a human-readable authorization memo.
- **Gemma 4 multimodal intake** — photo/voice capture, English/Swahili.

EXPLICITLY ROADMAP, NOT FAKED

- **On-device Gemma E2B/E4B** for true offline capture at low connectivity facilities, the client is already abstracted behind one interface for this swap.
- **Real DHIS2 / KEMSA LMIS / eCHIS integration** — today's data layer is a realistic mock shaped like the real adapter would be, not a placeholder.

WHY TRACK 3

Track 3 asks for stock monitoring, bed/staffing visibility, diagnostics availability, stock-out and demand forecasting, and redistribution recommendations that flag underperforming facilities. AfyaPulse covers all of it, and goes one step further than most stock dashboards: it doesn't just show the crisis, it computes the fix and lets you dispatch it.